

Tarea No. 1

Sección 1.4

1.1

a) cuánto km hay en 1.00 milla

$$1 \text{ mi} \times \frac{1.609 \text{ km}}{1 \text{ mi}} = \underline{1.609 \text{ km}}$$

b) cuánto pie hay en 1.00 Km

$$1 \text{ km} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{3.28084 \text{ ft}}{1 \text{ m}} = \underline{3,280.84 \text{ ft}}$$

1.3 ¿Cuántos nanosegundos tarda la luz en viajar 1 ft en el vacío?

$$\text{Vel. luz} = 300,000 \text{ Km/seg}$$

$$1 \text{ ft} \times \frac{0.30 \text{ m}}{1 \text{ ft}} = 0.30 \text{ m}$$

$$300,000 \frac{\text{km}}{\text{seg}} \times \frac{1000 \text{ m}}{1 \text{ km}} = 300,000,000 \text{ m/seg}$$

$$V = d/t \quad 1 \times 10^{-9} \text{ seg} \times \frac{1 \text{ seg}}{10^{-9} \text{ seg}} = \underline{1 \text{ nanosegundo}}$$

$$t = \frac{0.30 \text{ m}}{300,000,000 \text{ m/seg}} = 1 \times 10^{-9} \text{ seg}$$

$$1.5 \text{ Desplazamiento} = 327 \text{ in}^3$$

$$327 \text{ in}^3 \cdot \frac{(2.54 \text{ cm})^3}{(1 \text{ in})^3} = 5,358.57 \text{ cm}^3 \cdot \frac{1 \text{ L}}{1,000 \text{ cm}^3} = 5.35857 \text{ L}$$

1.7 ¿Cuántos años más tendrá usted dentro de 1.00 mil millones de segundos?

$$1,000,000,000 \text{ seg} \cdot \frac{1 \text{ min}}{60 \text{ seg}} \cdot \frac{1 \text{ hrs}}{60 \text{ min}} \cdot \frac{1 \text{ día}}{24 \text{ hrs}} \cdot \frac{1 \text{ año}}{365 \text{ días}} = 31.71 \text{ años}$$

1.9

a) mpg a Km/L

$$55 \frac{\text{mi}}{\text{g}} \cdot \frac{1.609 \text{ km}}{1 \text{ mi}} = 88.495 \frac{\text{km}}{\text{g}} \cdot \frac{0.264172 \text{ gal}}{1 \text{ L}} = 23.38 \text{ km/L}$$

b) Rendimiento = $\frac{\text{Dist}}{\text{red}}$

$$\text{Vol} = \frac{\text{Dist}}{\text{Rend}} = \frac{1,500 \text{ km}}{23.38 \frac{\text{km}}{\text{L}}} = 64.16 \text{ L}$$

$$\frac{64.16 \text{ L}}{45} = 1.42 \text{ veces deberá llenar el tanque}$$

9.11 Neptunio

$$mas = 60 \text{ Kg}$$

$$densidad = 19.5 \text{ g/cm}^3$$

$$Vol. esfera = \frac{4}{3} \pi r^3$$

$$60 \text{ Kg} \cdot \frac{1.000 \text{ g}}{1 \text{ Kg}} = 60,000 \text{ g}$$

$$Vol. esfera = \frac{60,000 \text{ g}}{19.5 \text{ g/cm}^3} = 3,076.92 \text{ cm}^3$$

$$r = \sqrt[3]{\frac{3}{4} \cdot \frac{3,076.92 \text{ cm}^3}{\pi}} = 9.02 = \underline{9 \text{ cm}}$$

Sección 1.7

1.27

Vector	Magnitud	Dirección	Comp X	Comp Y
$\vec{A}$	2.6 Km	90°	0	2.6 Km
$\vec{B}$	4 Km	0°	4 Km	0
$\vec{C}$	3.7 Km	45°	2.59 Km	2.59 Km
$\vec{R}$			6.59 Km	4.79 Km

$$\vec{C} = 3.7 \cos 45 = 2.59 \text{ Km}, \quad 3.7 \sin 45 = 2.59 \text{ Km}$$

$$R = \sqrt{(6.79)^2 + (4.79)^2}$$

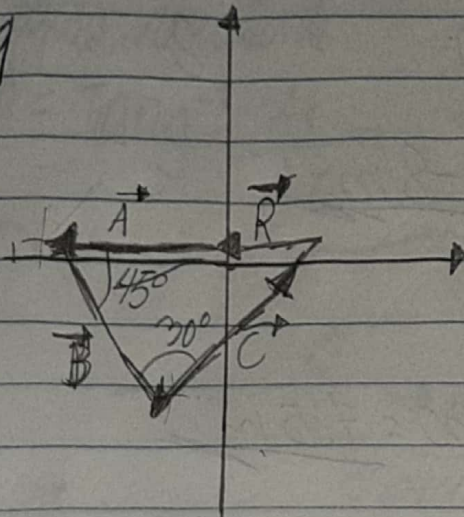
$$R = 7.83 \text{ km}$$

$$D = \theta = \tan^{-1} \left(\frac{4.79}{6.79} \right)$$

$$D = 37.73^\circ$$

$\vec{R} = 7.83 \text{ km}, 37.73^\circ$ este al norte

1.29



$\vec{A} = 180 \text{ m}$ al oeste

$\vec{B} = 210 \text{ m}$ al este del sur

$\vec{C} = 280 \text{ m}$ 30° al este del sur

Vect	Mag	Dirección	Comp X	Comp Y
$\vec{A}$	180 m	0	-180	0
$\vec{B}$	210 m	45°	148.49	-148.49
$\vec{C}$	280 m	30°	242.49	140
$\vec{R}$			210.98	-8.49

$$A = -180$$

$$B = 210 \cos 45 = 148.49$$

$$210 \sin 45 = 148.49$$

$$C = 280 \cos 30 = 242.49$$

$$280 \sin 30 = 140$$

$$R = \sqrt{(210.98)^2 + (-8.49)^2}$$

$$R = 211.75$$

$$D = \theta = \tan^{-1} \left(\frac{-8.49}{210.98} \right)$$

$$D = -2.30^\circ \text{ sur oeste}$$

Sección 1.8

1.31 Calcule las componentes x y y de los vectores $\vec{A}$, $\vec{B}$, $\vec{C}$, $\vec{D}$

$\vec{A} = 8\text{ m}$ 270° 0 8

x $8 \cos 90 = 0\text{ m}$ y $8 \sin 90 = -8\text{ m}$

$\vec{B} = 15\text{ m}$ 30°

x $15 \cos 30 = 12.99\text{ m}$ y $15 \sin 30 = 7.5\text{ m}$

$\vec{C} = 12\text{ m}$ 25° nor. este

x $12 \cos 25 = 10.88$ y $12 \sin 25 = 5.07$

$\vec{D} = 10\text{ m}$ 53° nor. este

$10 \cos 53 = 6.02\text{ m}$ $10 \sin 53 = 7.99$

1.33

a) ¿cuál es el componente x de $\vec{A}$?

Vect	$\vec{A}$
Mag	15.68
Dir	34°
x	-8.77
y	12.7 m

$h = \frac{12}{\cos 34} = 13.62$

b) $a = \sqrt{(15.68)^2 - (12)^2}$
 $a = 8.77$

7.35

a) $\vec{A} + \vec{B}$

Vect	Mag	Dir	Comp x	Comp y
A	8	270°	0	-8
B	15	30°	7.5	7.299
R			7.5	4.99

$$A = 8 \cos 270^\circ = 0$$

$$8 \sin 270^\circ = -8$$

$$B = 15 \cos 30^\circ = 7.299$$

$$15 \sin 30^\circ = 7.5$$

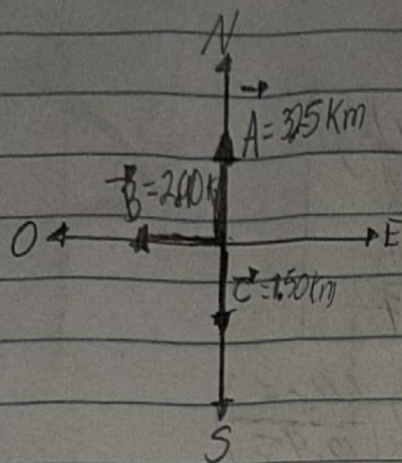
$$R = \sqrt{(7.5)^2 + (4.99)^2}$$

$$R = 9m$$

$$D = \tan^{-1} \left(\frac{4.99}{7.5} \right)$$

$$D = 33.69^\circ$$

1.37



Vect	Mag	Dir	Comp X	Comp Y
A	3.25 km	90°	0	3.25
B	2.90 km	180°	-2.9	0
C	1.5 km	270°	0	-1.5
R			-2.9	1.75

$$A = 3.25 \cos 90 = 0$$

$$3.25 \sin 90 = 3.25$$

$$B = 2.90 \cos 180 = -2.9$$

$$2.90 \sin 180 = 0$$

$$C = 1.5 \sin 270 = -1.5$$

$$1.5 \cos 270 = 0$$

$$R = \sqrt{(-2.9)^2 + (1.75)^2}$$

$$R = 3.39 \text{ km}$$

$$D = \tan^{-1}\left(\frac{1.75}{-2.9}\right)$$

$$D = +37.77^\circ \text{ norte de este}$$

1.39

a) $\vec{A} + \vec{B}$

Vect	Mag	Dir	Comp X	Comp Y
A	2.80 cm	60°	1.4	2.42
B	1.90 cm	300°	0.95	-1.65
R			2.35	0.77

$$A = 2.80 \cos 60 = 1.4$$

$$2.80 \sin 60 = 2.42$$

$$B = 1.90 \cos 300 = 0.95$$

$$1.90 \sin 300 = -1.65$$

$$R = \sqrt{(2.35)^2 + (0.77)^2} = 2.47 \text{ cm}$$

$$D = \tan^{-1}\left(\frac{0.77}{2.35}\right)$$

$$D = 18.14^\circ$$

b) $\vec{A} - \vec{B}$

Vect	Mag	Dir	Comp X	Comp Y
A	2.80 cm	60°	1.4	2.42
B	7.90 cm	60°	-0.95	7.65
R			0.45	4.07

$$R = \sqrt{(0.45)^2 + (4.07)^2}$$

$$R = 4.09 \text{ cm}$$

$$D = \tan^{-1} \left(\frac{4.07}{0.45} \right)$$

$$D = 83.69^\circ$$

c) $\vec{B} - \vec{A}$

Vect	Mag	Dir	Comp X	Comp Y
A	2.80 cm	60°	-1.4	-2.42
B	7.90 cm	60°	0.95	-7.65
R			-0.45	-4.07

$$R = \sqrt{(-0.45)^2 + (-4.07)^2}$$

$$R = 4.09 \text{ cm}$$

$$D = \tan^{-1} \left(\frac{-4.07}{-0.45} \right)$$

$$D = 83.7 + 180$$

$$D = 264^\circ$$

Problema 7.9

7.41	Vect	Mag	Dir	Comp X	Comp Y
	A	8	5	0	8
	B	15	30°	7.5	13
	C	12	25	10.9	5.1
	D	10	53	8	6.02

$$B = 15 \cos 30 = 13$$

$$15 \sin 30 = 7.5$$

$$C = 12 \cos 25 = 10.9$$

$$12 \sin 25 = 5.1$$

$$D = 10 \cos 53 = 6.02$$

$$10 \sin 53 = 8$$

$$\vec{A} = -8\hat{j}$$

$$\vec{B} = 7.5\hat{i} + 13\hat{j}$$

$$\vec{C} = (-10.9\hat{i}) + (-5.1\hat{j})$$

$$\vec{D} = (-6.02\hat{i}) + 8\hat{j}$$

1.43

a)	Vert	Mag	Dir	Comp X	Comp Y
A		3.60	70°	1.23	3.40
B		2.4	30°	-2.70	-1.20

$$A = 3.60 \cos 70 = 1.23$$

$$3.60 \sin 70 = 3.40$$

$$B = 2.4 \cos 30 = 2.70$$

$$2.4 \sin 30 = 1.20$$

$$\vec{A} = 1.23\hat{i} + 3.40\hat{j}$$

$$\vec{B} = (-2.70\hat{i}) + (-1.20\hat{j})$$

$$b) \vec{C} = 3.00\vec{A} - 4.00\vec{B}$$

$$C = 3 [(1.23\hat{i}) + (3.40\hat{j})] - 4 [(-2.70\hat{i}) + (-1.20\hat{j})]$$

$$C = (3.69\hat{i}) + (10.2\hat{j}) + (10.8\hat{i}) + (4.8\hat{j})$$

$$C = 14.5\hat{i} + 15.0\hat{j}$$

$$c) C = \sqrt{(14.5)^2 + (15.0)^2}$$

$$C = 20.7\text{ m}$$

$$\theta = \tan^{-1} \left(\frac{15.0}{14.5} \right)$$

$$\theta = 46.3^\circ$$

Sección 1.10

7.45	Vect	Mag	Dir	Comp _x	Comp _y
	A	8	5°	0	-8
	B	15	30°	7.5	12.99
	C	12	25°	-10.88	-5.07
	D	10	53°	6.02	-7.99

a) $\vec{A} \cdot \vec{B}$

A = 8 $(8)(15) \cos 130 = -104 \text{ m}^2$
 B = 15

b) $\vec{B} \cdot \vec{C}$

B = 15 $(15)(12) = 180 \cos 145 = -148 \text{ m}^2$
 C = 12
 $30 + 25 + 90 = 145$

c) $\vec{A} \cdot \vec{C}$

A = 8 $(8)(12) = 96 \cos 65 = 40.6 \text{ m}^2$
 C = 12
 $90 - 25 = 65$

Vect Limitaric

9.47 Calcule el ángulo entre estos pares de vectores

a) $A = -2\hat{i} + 6\hat{j}$ y $B = 2\hat{i} - 3\hat{j}$

$$A \cdot B = [(-2)(2)] + [(6)(-3)]$$

$$A \cdot B = -4 + (-18) = -22$$

$$A = \sqrt{(-2)^2 + (6)^2}$$

$$A = 6.32$$

$$B = \sqrt{(2)^2 + (-3)^2}$$

$$B = 3.61$$

$$AB = 6.32 \times 3.61 = 22.8152$$

$$\theta = \cos^{-1}\left(\frac{-22}{22.8152}\right) = 165^\circ$$

b) $A = 3\hat{i} + 5\hat{j}$ y $B = 10\hat{i} + 6\hat{j}$

$$A \cdot B = [(3)(10)] + [(5)(6)]$$

$$A \cdot B = 30 + 30 = 60$$

$$A = \sqrt{(3)^2 + (5)^2}$$

$$A = 5.83$$

$$B = \sqrt{(10)^2 + (6)^2}$$

$$B = 11.66$$

$$AB = 5.83 \times 11.66 = 67.9778$$

$$\theta = \cos^{-1}\left(\frac{60}{67.9778}\right) = 28.01^\circ$$

c) $\vec{A} = -4\hat{i} + 2\hat{j}$ y $\vec{B} = 7\hat{i} + 14\hat{j}$

$$\vec{A} \cdot \vec{B} = [(-4)(7)] + [(2)(14)]$$

$$\vec{A} \cdot \vec{B} = -28 + 28 = 0$$

$$A = \sqrt{(-4)^2 + (2)^2}$$

$$A = 4.47$$

$$B = \sqrt{(7)^2 + (14)^2}$$

$$B = 15.65$$

$$AB = 4.47 \times 15.65 = 70.0$$

$$\theta = \cos^{-1}\left(\frac{0}{70.0}\right) = 90^\circ$$

7.49a)	Vect	Mag	Dir	Comp x	Comp y
A		8	5°	0	-8
D		10	53°	6.02	-7.99

$$\vec{A} \times \vec{D} = (8)(10) = 80 \cos 37^\circ = 63.90 \text{ m}^2 \text{ R}$$

$$90 - 53 = 37$$

b) $\vec{D} \times \vec{A}$

$$(10)(8) = 80 \cos 37^\circ = 63.90 \text{ m}^2 \text{ R}$$

$$90 - 53 = 37$$

1.51 a) $\vec{A} \cdot \vec{B}$

$$\vec{A} \cdot \vec{B} = (3.6)(2.4) \cos 140 = -6.62 \text{ m}^2$$

$$90 + 30 + 20 = 140$$

b) $\vec{A} \times \vec{B}$

$$\vec{A} \times \vec{B} = (3.6)(2.4) \sin 140 = 5.55 \text{ m}^2$$

$$90 + 30 + 20 = 140$$

Problemas

1.65

$\vec{A} = 25^\circ$ este del norte

$\vec{B} = 56^\circ$, N

$c = 32$, 0

$$A = A_x = 19.07$$

$$\cos 34.723$$

$$A_y = -12.86$$

$$\sin 34.723$$

$$B = B_x = 0$$

$$y = 46$$

$$C_x = 32$$

$$y = 0$$

Vect	x	y
A	19.07	12.86
B	0	46
C	32	0

$$\sqrt{(19.07)^2 + (33.74)^2} = 60.9 \text{ m}$$

$$\theta = \tan^{-1} \left(\frac{33.74}{19.07} \right) = 33^\circ$$

$$\vec{A} + \vec{B} + \vec{C} = 51.07 \uparrow + 33.74 \uparrow$$

7.67

$\vec{A} =$
 $\vec{B} = 6.40 \text{ cm}, 63^\circ$
 $\vec{C} = 6.40, 22^\circ$

Vect	Mag	Dir	Comp X	Comp Y
A			3.3	8.84
B	6.40	63	-5.70	2.97
C	6.40	22	2.40	5.93

$B = 6.40 \cos 63 = 2.97$
 $6.40 \sin 63 = 5.70$

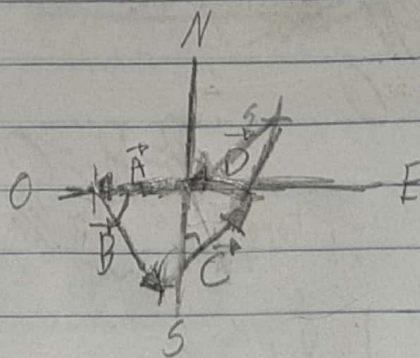
$C = 6.40 \cos 22 = 5.93$
 $6.40 \sin 22 = 2.40$

$\sqrt{(8.84)^2 + (-3.3)^2}$
 8.75 cm

$\theta = \tan^{-1} \left(\frac{8.84}{3.3} \right)$
 $\theta = 69.5^\circ$

7.69

$\vec{A} = 180 \text{ m}, \text{ al oeste}$
 $\vec{B} = 210 \text{ m}, 45^\circ \text{ este del sur}$
 $\vec{C} = 280 \text{ m}, 30^\circ \text{ este del norte}$
 $\vec{D} = ?$



Vect	Mag	Dir	Comp X	Comp Y	
A	180	oeste	-180	0	$B = 210 \cos 45 = 148.49$
B	210	45°	148.5	-148.5	$210 \sin 45 = 148.49$
C	280	30°	140	242.5	$C = 280 \cos 30 = 242.5$
D			-108.5	-94	$280 \sin 30 = 140$
R	0	0	0	0	

$$D = \sqrt{(-108.5)^2 + (-94)^2}$$

$$= 144 \text{ m}$$

$$\theta = \tan^{-1} \left(\frac{-94}{-108.5} \right)$$

$$\theta = 41^\circ \text{ sur norte}$$

7.71

$\vec{A} = 480 \text{ N}$, hacia adelante
 $\vec{B} = 513 \text{ N}$, 32.4°

Vect	Magn	Dir	Comp X	Comp Y
A	480	0	480	0
B	513	32.4°	433.14	274.9
R			913.14	274.9

$$B = 513 \cos 32.4^\circ = 433.14$$

$$513 \sin 32.4^\circ = 274.9$$

$$D = \sqrt{(913.14)^2 + (274.9)^2}$$

$$= 954 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{274.9}{913.14} \right)$$

$$\theta = 16.80^\circ$$

7.73

$$F_{\text{resultante}} = \sqrt{F_a^2 + F_b^2 + 2 F_a F_b \cos \theta}$$

$$F_r = 5.6 \text{ N}$$

$$F_a = F_b$$

$$32 + 32 = 64^\circ$$

$$5.6^2 = 2 F_a^2 + F_a^2 \cos 64$$

$$31.36 = 2 F_a^2 (1 + 0.44)$$

$$2 F_a^2 = 21.80$$

$$F_a^2 = 10.9$$

$$F_a = 3.30 \text{ N}$$

7.75 Equilibrium

$$100 \text{ N} \cos 70^\circ - F \cos \theta = 0 \quad F \cos \theta = 34.2 \text{ N}$$

$$100 \text{ N} \sin 70^\circ + F \sin \theta - 124 \text{ N} = 0 \quad F \sin \theta = 30 \text{ N}$$

$$\frac{F \sin \theta}{F \cos \theta} = \tan \theta = \frac{30}{34.2} = 0.877$$

$$F = \frac{34.2 \text{ N}}{\cos 47.26^\circ} = 45.5 \text{ N}$$

$$\theta = 47.26^\circ$$

Vect	Mag	Dir	Comp X	Comp Y	
A	100	70	34.2	94	A = 100 cos 70 = 34.2 100 sin 70 = 94
B	124	90	0	-724	B = 124 cos 90 = 0 124 sin 90 = 124
R			34.2	-30	

$$R = \sqrt{(34.2)^2 + (-30)^2} \quad \theta = \tan^{-1} \left(\frac{-30}{34.2} \right)$$

$$R = 46 \text{ N}$$

$$\theta = -47^\circ$$

$$\theta = 180 - 47 = 133^\circ$$

7.77

$$C = \sqrt{(32)^2 + (27)^2 - 2(32)(27)\cos 60}$$

$$C = \sqrt{(1024) + (441) - 672}$$

$$C = 28.76 \text{ m}$$

1.79 Humeros y músculos

$$\vec{A} = 232\text{ N}, 43^\circ$$

X	Y
158.22 N	169.67 N

$$232 \text{ N} \cos 43^\circ = 169.67 \text{ N}$$

$$232 \text{ N} \sin 43^\circ = 158.22 \text{ N}$$

$$Y = -25\text{ N} - 112\text{ N} + 169 = 32\text{ N}$$

$$R = \sqrt{(158.22)^2 + (-32)^2}$$

$$R = 161.4$$

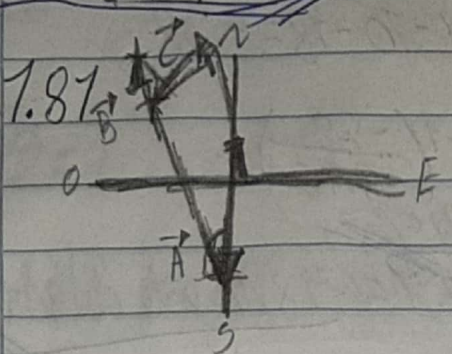
$$X = 158.22 \text{ N}$$

$$\theta = \tan^{-1}\left(\frac{-32}{158.22}\right)$$

$$\theta = 77.8^\circ \approx 12^\circ$$

$$C = 158\hat{i} - 33\hat{j}$$

$$C = 161\text{ N } 13^\circ$$



$$\vec{A} = 825\text{ m, al sur}$$

$$\vec{B} = 1.25\text{ Km, } 30^\circ \text{ oeste del norte}$$

$$\vec{C} = 1\text{ Km, } 40^\circ \text{ norte del este}$$

$$\vec{D} = ?$$

Vect	Mag	Dir	Comp X	Comp Y
A	0.825 Km	nur	0	-0.825
B	1.25 Km	30°	-1.08	0.625
C	1 Km	40°	0.766	-0.643
D			-0.314	-0.843

$$B = 1.25 \cos 30^\circ = 1.08$$

$$1.25 \sin 30^\circ = 0.625$$

$$C = 1 \cos 40^\circ = 0.766$$

$$1 \sin 40^\circ = 0.643$$

$$D = \sqrt{(-0.314)^2 + (-0.843)^2}$$

$$D = 0.911 \text{ Km} \approx 911\text{ m}$$

$$\theta = \tan^{-1}\left(\frac{-0.843}{-0.314}\right)$$

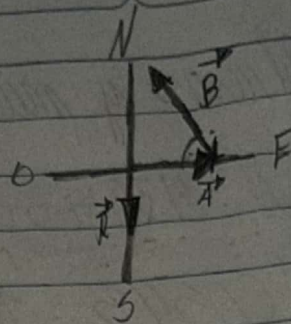
$$\theta = 69.57^\circ$$

m a Km

$$\frac{825\text{ m}}{1000} = 0.825\text{ Km}$$

1.83

$A = 72\text{m}$, este
 $B = 28\text{m}$, 50° este del norte
 $R = 10\text{m}$, Sur
 $C = ?$



Vect.	Mag	Dir	Comp X	Comp Y
A	72	este	72	0
B	28	50°	-27.45	18
C			9.45	-28
R	10	sur	0	-10

$$B = 28 \cos 50 = 18$$

$$28 \sin 50 = 27.45$$

Comp X

$$72 - 27.45 + X = 0$$

$$-9.45 + X = 0$$

$$X = 9.45$$

Comp Y

$$18 + Y = -10$$

$$Y = -10 - 18 = -28$$

$$D = \sqrt{(9.45)^2 + (-28)^2}$$

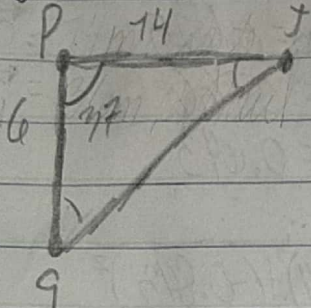
$$D = 29.5 \text{ m}$$

$$\theta = \tan^{-1}\left(\frac{-28}{9.45}\right)$$

$$\theta = 71.4^\circ$$

$$\theta = 90 - 71.4 = 18.6 \text{ este del sur}$$

1.85



$$JQ = 74^2 + 36^2 - (2 \cdot 74 \cdot 36) \cos 37$$

$$JQ = 26.2 \text{ m}$$

$$\theta = \left(\frac{36^2 - 74^2 - 2(6.2)}{-(72 \cdot 74 \cdot 26.2)} \right) \cos^{-1}$$

$$\theta = 124.37^\circ$$

$$190 - 124.37 = 65.63^\circ \text{ sur este}$$

1.87

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$|\vec{A} \times \vec{B}| = AB \sin \theta$$

$$\tan \theta = \frac{9}{6} = 1.5 \Rightarrow \theta = 56.3^\circ$$

$$\vec{A} \cdot \vec{B} = AB \cos \theta = -6$$

$$|\vec{A} \times \vec{B}| = AB \sin \theta = 9$$

1.89

$$\vec{A} \cdot \vec{B} = AB \cos \theta = 90 \text{ m}^2$$

$$\cos \theta = \frac{90 \text{ m}^2}{AB} = \frac{90 \text{ m}^2}{(12)(16)} = \frac{90 \text{ m}^2}{192} = 0.4688$$

$$\theta = 62.05^\circ$$

$$|\vec{A} \times \vec{B}| = AB \sin \theta = (12 \text{ m})(16 \text{ m}) \sin 62.05^\circ = 170 \text{ m}^2$$

1.91

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB}$$

$$\vec{A} = \vec{k}$$

$$A = 1$$

$$B = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\vec{B} = i + j + k$$

$$\vec{A} \cdot \vec{B} = k(i + j + k) = 1$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{1}{\sqrt{3}} = 54.1^\circ$$

$$\vec{A} = i + j + k$$

$$\vec{B} = i + k$$

$$A = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$B = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\vec{A} \cdot \vec{B} = (1 + j + k) \cdot (1 + k) = 1 + 1 = 2$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{2}{\sqrt{3} \cdot \sqrt{2}} = \frac{2}{\sqrt{6}} \quad \theta = 35.3^\circ$$

1.93

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

$$\theta = 79^\circ$$

$$B = \frac{\vec{A} \cdot \vec{B}}{A \cos \theta} = \frac{48 \text{ m}^2}{9 \cos 79^\circ} = 27.95 \approx 28 \text{ m}$$

1.95

Vert	$\uparrow$	$\uparrow$	
A	5	-6.5	
B	-3.5	7	= 15

$$A \cdot C = 0 \quad (5; -6.5) \cdot (C_x; C_y) = 0$$

$$5C_x - 6.5C_y = 0$$

$$5C_x = 6.5C_y$$

$$C_x = 6.5C_y / 5 = 1.3C_y$$

$$C_x = 1.3(-6.1)$$

$$C_x = -8$$

$$3.5C_x - 7C_y = 15$$

$$3.5(1.3C_y) - 7C_y = 15$$

$$4.55C_y - 7C_y = 15$$

$$-2.45C_y = 15$$

$$C_y = 15 / -2.45 = -6.1C_y$$